

Modeling a Multi-Provider Vehicle Retail Ecosystem for Interoperable Incentive Reasoning

Project Codename: Retail-Mesh-Intelligence

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ABSTRACT

Parent automotive companies operate through networks of dealers, rental agencies, and service providers, each selling, renting, and maintaining vehicles on the parent company's behalf and each rewarded through its own performance-based incentive program. Today, providers independently maintain their own records of sales, rentals, and services, and these records remain disconnected even when they concern the same vehicle or customer, creating a gap in understanding how events across providers relate. We define the problem of representing providers, vehicles, customers, and their sales, rental, and service transactions, together with the performance metrics and incentive rules that depend on them, in a shared, machine-understandable format that is globally accessible and interoperable, so that humans, software applications, and large language models can consume, integrate, query, and reason over the vehicle retail ecosystem. The codebase, installation instructions, and software artifacts will be made available under the cc-by-nc-sa-4.0 license at https://github.com/**.

KEYWORDS

Knowledge Graph, Vehicle Retail, Linked Data, Semantic Interoperability, Incentive Reasoning

1 INTRODUCTION

Vehicle ownership today rarely involves a single organization. Parent automotive groups operate through networks of dealers, rental agencies, and service centers that independently sell, rent, and maintain vehicles under one brand, earning rewards through loyalty programs tied to sales volume, rental usage, and service quality [2]. Rewarding this network fairly requires linking events across providers: a vehicle may be sold by one dealer, rented from another, and serviced by a third, all involving the same customer, yet these events are usually recorded independently and never connected.

Because each provider uses its own schema and identifiers, there is no single view of a vehicle or customer across systems. The relationships between providers and transaction types, such as whether a customer completed a full journey, are rarely made explicit, making it difficult to determine whether a provider earned a reward. This project addresses the problem of representing providers, vehicles, customers, and their sales, rental, and service transactions, together with the performance metrics and incentive rules that depend on them, in a globally accessible, interoperable format that humans,

applications, and computational agents, including LLMs, can use to understand, query, and reason about the vehicle retail ecosystem.

2 MOTIVATION

Such a representation could help integrate different provider systems using different schemas and identifiers for the same vehicles and customers [1], enable a connected view of a customer's journey across sales, rentals, and service from multiple providers [3], and give a chatbot or analytics tool a basis to explain why a provider did or did not earn a reward, making the system more transparent and fair [4].

2.1 Motivating Example

Imagine a customer buys a car from Dealer A. While it is being serviced, they rent a replacement from Provider B, and the car is repaired at Provider C. The parent company's loyalty program gives bonuses to providers whose customers complete a full sale-and-service cycle. To know whether Dealer A qualifies, three events must be connected: the sale, the rental, and the service, all involving the same customer and vehicle. Current dealer-management and CRM systems handle each transaction well on its own [3], but do not expose the links between them in a queryable form, so there is no way to trace the journey or calculate incentives. Our model aims to make these connections visible and usable.

3 TEAM JUSTIFICATION

Our team consists of three members with skills in database design, semantic web technologies, and applied artificial intelligence, mapping directly to the phases of the project: entity-relationship modeling and SQL design, RDF/OWL representation, and evaluation. Each member contributes to every milestone, and we track contributions using a shared Git repository and commit history.

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